

9.2.4: Similarity and Similitude

One of dimensional analysis is the key point is the concept that the solution can be obtained by conducting experiments on similar but not identical systems. The analysis here suggests and demonstrates that the solution is based on several dimensionless numbers. Hence, constructing experiments of the situation where the same dimensionless parameters obtains could, in theory, yield a solution to problem at hand. Thus, knowing what are dimensionless parameters should provide the knowledge of constructing the experiments. In this section deals with these similarities which in the literature some refer as analogy or similitude. It is hard to obtain complete similarity. Hence, there is discussion how similar the model is to the prototype. It is common to differentiate between three kinds of similarities: geometric, kinetics, and dynamic. This characterization started because historical reasons and it, some times, has merit especially when applying Buckingham's method. In Nusselt's method this differentiation is less important.

Geometric Similarity

One of the logical part of dimensional analysis is how the experiences should be similar to actual body they are supposed to represent. This logical conclusion is an add-on and this author is not aware of any proof to this requirement based on Buckingham's methods. Ironically, this conclusion is based on Nusselt's method which calls for the same dimensionless boundary conditions. Again, Nusselt's method, sometimes or even often, requires similarity because the requirements to the boundary conditions. Here this postulated idea is adapted. Under this idea the prototype area has to be square of the actual model or

$$\left[\frac{A_p}{A_m} \right] = \left(\frac{l_p}{l_m} \right)^2$$

where l_1 and l_2 are the typical dimensions in two different directions and subscript p refers to the prototype and m to the model. Under the same argument the volumes change with the cubes of lengths. In some situations, the model faces inability to match two or more dimensionless parameters. In that case, the solution is to sacrifice the geometric similarity to minimize the undesirable effects. For example, river modeling requires to distort vertical scales to eliminate the influence of surface tension or bed roughness or sedimentation.

Kinematic Similarity

The perfect kinetics similarity is obtained when there are geometrical similarity and the motions of the fluid above the objects are the same. If this similarity is not possible, then the desire to achieve a motion "picture" which is characterized by ratios of corresponding velocities and accelerations is the same throughout the actual flow field. It is common in the literature, to discuss the situations where the model and prototype are similar but the velocities are different by a different scaling factor. The geometrical similarity aside the shapes and counters of the object it also can requires surface roughness and erosion of surfaces of mobile surfaces or sedimentation of particles surface tensions. These impose demands require a minimum on the friction velocity. In some cases the minimum velocity can be $U_{min} = \sqrt{\tau_w / \rho}$. For example, there is no way achieve low Reynolds number with thin film flow.

Dynamics Similarity

The dynamic similarity has many confusing and conflicting definitions in the literature. Here this term refers to similarity of the forces. It follows, based on Newton's second law, that this requires similarity in the accelerations and masses between the model and prototype. It was shown that the solution is a function of several typical dimensionless parameters. One of such dimensionless parameter is the Froude number. The solution for the model and the prototype are the same, since both cases have the same Froude number. Hence it can be written that

$$\left(\frac{U^2}{g \ell} \right)_m = \left(\frac{U^2}{g \ell} \right)_p \quad (9.2.4.1)$$

It can be noticed that $t \sim \ell / U$ thus equation (46) can be written as

$$\left(\frac{U}{g t} \right)_m = \left(\frac{U}{g t} \right)_p \quad (9.2.4.2)$$

and noticing that $a \propto U/t$

$$\left(\frac{a}{g}\right)_m = \left(\frac{a}{g}\right)_p \quad (9.2.4.3)$$

and $a \propto F/m$ and $m = \rho \ell^3$ hence $a = F/\rho \ell^3$. Substituting into equation (48) yields

$$\left(\frac{F}{\rho \ell^3}\right)_m = \left(\frac{F}{\rho \ell^3}\right)_p \Rightarrow \frac{F_p}{F_m} = \frac{(\rho \ell^3)_p}{(\rho \ell^3)_m} \quad (9.2.4.4)$$

In this manipulation, it was shown that the ratio of the forces in the model and forces in the prototype is related to ratio of the dimensions and the density of the same systems. While in Buckingham's methods these hand waiving are not precise, the fact remains that there is a strong correlation between these forces. The above analysis was dealing with the forces related to gravity. A discussion about force related the viscous forces is similar and is presented for the completeness. The Reynolds numbers is a common part of Navier–Stokes equations and if the solution of the prototype and for model to be same, the Reynolds numbers have to be same.

$$Re_m = Re_p \Rightarrow \left(\frac{\rho U \ell}{\mu}\right)_m = \left(\frac{\rho U \ell}{\mu}\right)_p \quad (9.2.4.5)$$

Utilizing the relationship $U \propto \ell/t$ transforms equation (??) into

$$\left(\frac{\rho \ell^2}{\mu t}\right)_m = \left(\frac{\rho \ell^2}{\mu t}\right)_p \quad (9.2.4.6)$$

multiplying by the length on both side of the fraction by ℓU as

$$\left(\frac{\rho \ell^3 U}{\mu t \ell U}\right)_m = \left(\frac{\rho \ell^3 U}{\mu t \ell U}\right)_p \Rightarrow \frac{(\rho \ell^3 U/t)_m}{(\rho \ell^3 U/t)_p} = \frac{(\mu \ell U)_m}{(\mu \ell U)_p} \quad (9.2.4.7)$$

Noticing that U/t is the acceleration and $\rho \ell$ is the mass thus the forces on the right hand side are proportional if the Re number are the same. In this analysis/discussion, it is assumed that a linear relationship exist. However, the Navier–Stokes equations are not linear and hence this assumption is excessive and this assumption can produce another source of inaccuracy. While this explanation is a poor practice for the real world, it common to provide questions in exams and other tests on this issue. This section is provide to this purpose.

Example 9.9

The liquid height rises in a tube due to the surface tension, σ is h . Assume that this height is a function of the body force (gravity, g), fluid density, ρ , radius, r , and the contact angle θ . Using Buckingham's theorem develop the relationship of the parameters. In experimental with a diameter 0.001 [m] and surface tension of 73 milli-Newtons/meter and contact angle of 75° a height is 0.01 [m] was obtained. In another situation, the surface tension is 146 milli-Newtons/meter, the diameter is 0.02 [m] and the contact angle and density remain the same. Estimate the height.

Solution

It was given that the height is a function of several parameters such

$$h = f(\sigma, \rho, g, \theta, r) \quad (9.2.4.8)$$

There are 6 parameters in the problem and the 3 basic parameters [L , M , t]. Thus the number of dimensionless groups is (6-3=3). In Buckingham's methods it is either that the angle isn't considered or the angle is dimensionless group by itself. Five parameters are left to form the next two dimensionless groups. One technique that was suggested is the possibility to use three parameters which contain the basic parameters [M , L , t] and with them form a new group with each of the left over parameters. In this case, density, ρ for [M] and d for [L] and gravity, g for time [t].

For the surface tension, σ it becomes

$$\left[\frac{\rho}{M L^{-3}}\right]^a \left[\frac{r}{L}\right]^b \left[\frac{g}{L t^{-2}}\right]^c \left[\frac{\sigma}{M t^{-2}}\right]^1 = M^0 L^0 t^0 \quad (9.2.4.9)$$

Equation (54) leads to three equations which are

$$\begin{array}{rcl} \text{Mass, } M & a + 1 = & 0 \\ \text{Length, } L & -3a + b + c = & 0 \\ \text{time, } t & -2c - 2 = & 0 \end{array} \quad (9.2.4.10)$$

the solution is $a = -1$ $b = -2$ $c = -1$ Thus the dimensionless group is $\frac{\sigma}{\rho r^2 g}$. The third group obtained under the same procedure to be h/r . In the second part the calculations for the estimated of height based on the new ratios. From the above analysis the functional dependency can be written as

$$\frac{h}{d} = f\left(\frac{\sigma}{\rho r^2 g}, \theta\right) \quad (9.2.4.11)$$

which leads to the same angle and the same dimensional number. Hence,

$$\frac{h_1}{d_1} = \frac{h_2}{d_2} = f\left(\frac{\sigma}{\rho r^2 g}, \theta\right) \quad (9.2.4.12)$$

Since the dimensionless parameters remain the same, the ratio of height and radius must be remain the same. Hence,

$$h_2 = \frac{h_1 d_2}{d_1} = \frac{0.01 \times 0.002}{0.001} = 0.002 \quad (9.2.4.13)$$

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